THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

Convex Optimization for Machine Learning

COMP6704 Group 10: L2-Regularized Logistic Regression

LIU Xuyang WANG Zeqing WU Aoqi ZHAO Jixin
Department of Computing, The Hong Kong Polytechnic University
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Agenda and speakers

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30-minute presentation, followed by 10 minutes of Q&A.

One training problem, four solvers



Our question

How much work does each solver need to train the same model?

- Same model, data and starting point.
- Same L2 penalty and stopping target.
- Compare training error, time and test accuracy.

$$\min_{w} F(w) \quad \text{using GD, Nesterov, FISTA or ADMM.}$$

This problem has one best set of weights.

L2-regularized logistic regression



$$\min_{w \in \mathbb{R}^d} F(w), \qquad F(w) = f(w) + g(w),$$

$$f(w) = \frac{1}{n} \sum_{i=1}^{n} \log\left(1 + e^{-y_i x_i^\top w}\right), \qquad g(w) = \frac{\lambda}{2} \|w\|_2^2.$$

- x_i: input features. y_i: class label, -1 or $+1$.
- n: training examples. w: the d model weights.
- Loss measures prediction errors. L2 discourages large weights.
- $\lambda = 0.01$. The penalty includes the bias.

Implementation: `problem.py`, `data.py`.

Predictions and the gradient

Put the training examples in the rows of $X \in \mathbb{R}^{n \times d}$.

$$p_i = \sigma(x_i^\top w), \qquad \sigma(a) = \frac{1}{1 + e^{-a}}, \qquad b_i = \frac{y_i + 1}{2}.$$

The gradient tells us how to change the weights

$$\nabla F(w) = \frac{1}{n} X^\top (p - b) + \lambda w.$$

- Predict $+1$ when $x_i^\top w \geq 0$ and -1 otherwise.
- One gradient needs $O(nd)$ work on dense data.
- Train with F. Check predictions on separate test data.

Why the minimizer is unique

Let $D = \operatorname{diag}(p_i(1 - p_i))$. Since $0 \preceq D \preceq I/4$,

$$\nabla^2 F(w) = \frac{1}{n} X^\top D X + \lambda I, \qquad \lambda I \preceq \nabla^2 F(w) \preceq \left(\frac{\|X\|_2^2}{4n} + \lambda \right) I.$$

Two bounds help us choose the steps

$$\mu = \lambda > 0, \qquad L_f = \frac{\|X\|_2^2}{4n}, \qquad L = L_f + \lambda.$$

- The positive L2 penalty gives one best answer, $w^\star$.
- This property is called **strong convexity**.
- L limits how fast the gradient changes.

What counts as convergence?



We stop when the full gradient is small enough:

$$\|\nabla F(w_k)\|_2 \le \varepsilon, \qquad \varepsilon = 10^{-6}.$$

This also bounds the gap from the best objective value:

$$0 \le F(w_k) - F(w^\star) \le \frac{\|\nabla F(w_k)\|_2^2}{2\lambda} \le 5 \times 10^{-11} \quad \text{when the check passes.}$$

- Stop after 400 updates even if the target is not met.
- Matching test accuracy does not mean training is finished.
- ADMM also checks its two copies and inner solve.

Four implementations

Solver	Main idea	Code owner
GD	Follow the downhill direction	WANG Zeqing
Nesterov	Look ahead, then step	LIU Xuyang
FISTA	Step, then shrink the weights	WU Aoqi
ADMM	Update two copies, then match them	ZHAO Jixin

All four minimize $F(w) = f(w) + \frac{\lambda}{2}\|w\|_2^2$.

Modules: `algorithms/gd.py`, `nesterov.py`, `fista.py`, `admm.py`.

Gradient descent

$$w_{k+1} = w_k - \eta_k \nabla F(w_k), \qquad \eta_k \text{ starts at } 1/L.$$

- Start with all weights at zero.
- If the loss plus penalty does not fall enough, halve the step:

$$F(w_k - \eta_k \nabla F(w_k)) \le F(w_k) - 10^{-4} \eta_k \|\nabla F(w_k)\|_2^2.$$

How the error bound shrinks

With step $1/L$, the bound shrinks by a fixed factor each update:

$$F(w_k) - F(w^\star) \le (1 - \mu/L)^k \, [F(w_0) - F(w^\star)].$$

Nesterov accelerated gradient

Initialize $w_0 = v_0 = 0$ and $t_0 = 1$. At update k:

$$w_{k+1} = v_k - \frac{1}{L}\nabla F(v_k),$$
$$t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2},$$
$$v_{k+1} = w_{k+1} + \frac{t_k - 1}{t_{k+1}}(w_{k+1} - w_k).$$

- v_k looks ahead using the last two weight vectors.
- The objective-error bound falls as $O(1/k^2)$.
- We use the basic schedule, with no restart.
- Some updates can increase the objective.

FISTA: gradient plus proximal step



FISTA handles the loss f and penalty g in separate steps.

$$\text{prox}_{\eta g}(a) = \arg\min_w \left\{ g(w) + \frac{\|w-a\|_2^2}{2\eta} \right\} = \frac{a}{1+\eta\lambda}.$$

Take a loss step, then shrink the weights

$$w_{k+1} = \frac{v_k - \eta \nabla f(v_k)}{1+\eta\lambda}, \qquad \eta = 1/L_f.$$

- Use the same look-ahead rule as Nesterov.
- Apply the L2 penalty once, through the shrink step.
- The objective-error bound falls as $O(1/k^2)$.

Beck and Teboulle, SIAM Journal on Imaging Sciences, 2009.

Why Nesterov and FISTA overlap here



For this L2 penalty and the default step sizes,

$$\begin{aligned}\frac{v - \nabla f(v)/L_f}{1 + \lambda/L_f} &= \frac{L_f v - \nabla f(v)}{L_f + \lambda} \\ &= v - \frac{\nabla f(v) + \lambda v}{L_f + \lambda} \\ &= v - \frac{1}{L}\nabla F(v).\end{aligned}$$

- Here, both formulas give the **same update**.
- Same start and momentum mean the curves overlap.
- Different penalties or step sizes can change this.

ADMM: split the loss and penalty



ADMM gives the loss and penalty their own weight vectors:

$$\min_{w,z} f(w) + \frac{\lambda}{2}\|z\|_2^2 \quad \text{subject to } w = z.$$

The variable u tracks their disagreement. We use $\rho = 1$.

$$\begin{aligned} w_{k+1} &= \arg\min_{w} \left\{ f(w) + \frac{\rho}{2}\|w - z_k + u_k\|_2^2 \right\}, \\ z_{k+1} &= \frac{\rho}{\rho + \lambda}(w_{k+1} + u_k), \\ u_{k+1} &= u_k + w_{k+1} - z_{k+1}. \end{aligned}$$

- Updating w needs an inner Newton solve.
- Updating z uses a direct formula.

Boyd et al., Foundations and Trends in Machine Learning, 2011.

Work per update and stopping

Method	Work on dense data	Extra work
GD	$O(nd)$	Check the step
Nesterov / FISTA	$O(nd)$	Check the full gradient
ADMM	$O(nd^2 + d^3)$ per Newton step	May need several inner steps

ADMM allows 50 Newton steps per outer update. Its inner gradient tolerance is 10^{-10}.

$$r_{k+1} = w_{k+1} - z_{k+1}, \qquad s_{k+1} = \rho(z_{k+1} - z_k).$$

The same update count can mean very different amounts of work.

Experimental setup

Dataset	Train	Test	Parameters, including bias
Synthetic	1,500	500	51
Breast cancer	426	143	31

- 75/25 split, seed 0. Keep class proportions similar.
- Learn feature scaling from training data only.
- Same start, $\lambda = 0.01$, 400 updates and 10^{-6} target.
- Recheck final scores with the shared code.
- Time the solver call, including setup and logging.

Saved runs: `results/summary.json` and `results/breast_cancer/summary.json`.

Synthetic: all four pass the gradient check

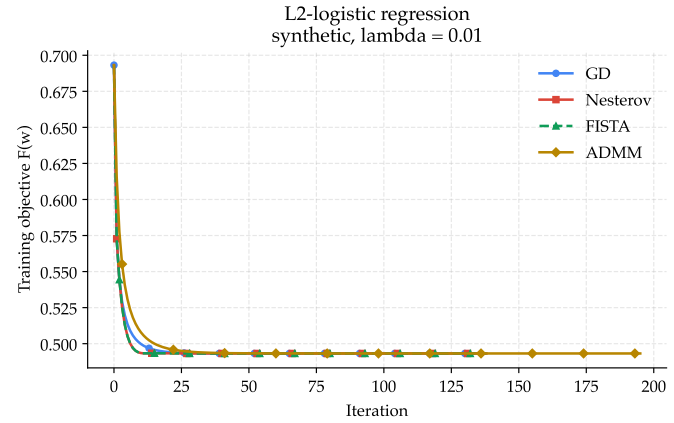
Method	Final F	$\|\nabla F\|_2$	Updates	Wall, ms	Test acc.
GD	0.4932140405	9.81×10^{-7}	136	7.08	73.80%
Nesterov	0.4932140405	8.69×10^{-7}	138	8.52	73.80%
FISTA	0.4932140405	8.69×10^{-7}	138	8.11	73.80%
ADMM	0.4932140405	9.68×10^{-7}	195	54.90	73.80%

- All reach $F(w) \approx 0.4932140405$ and 73.80% test accuracy.
- GD takes 136 updates; Nesterov and FISTA take 138.
- ADMM takes 195 outer updates and passes its extra checks.

GD reaches the target in slightly fewer updates here.

One saved run, seed 0. Repeat timings before ranking small differences.

Synthetic objective versus iteration



Raw training objective $F(w_k)$; iteration zero is initialization.

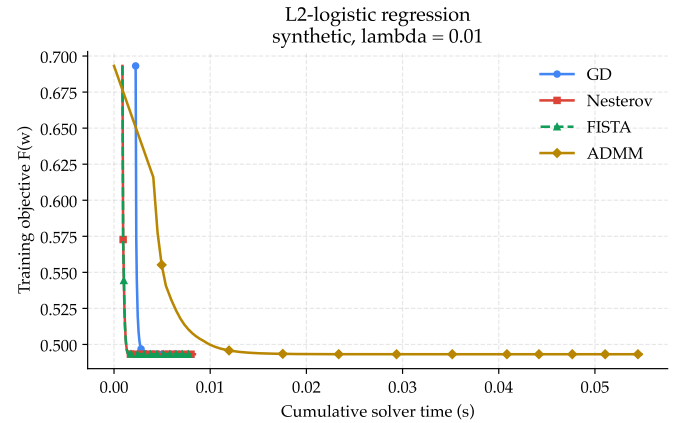
$F(w_0) = \log 2.$

All curves approach the same objective.

Nesterov and FISTA overlap.

This scale hides tiny differences near the end.

Synthetic objective versus time



ADMM's inner solves take more time here.

The other three finish close together.

We need repeated runs to compare small time differences.

Curves use internal cumulative time; table wall time covers the entire solver call.

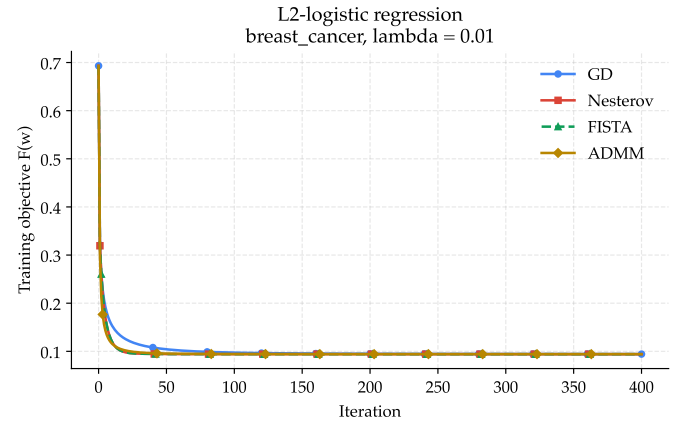
Breast cancer: the budget ends first

Method	Final F	$\|\nabla F\|_2$	Updates	Wall, ms	Test acc.
GD	0.0942636788	1.67×10^{-3}	400	7.96	96.50%
Nesterov	0.0941673920	4.23×10^{-5}	400	13.23	96.50%
FISTA	0.0941673920	4.23×10^{-5}	400	12.64	96.50%
ADMM	0.0941674245	4.50×10^{-5}	400	54.91	96.50%

- All use 400 updates. **None reaches the 10^{-6} target.**
- Test accuracy is 96.50% for all, but gradient sizes differ.
- Nesterov and FISTA have lower objective values than GD here.

Same seed, λ and update budget. The separate 1,000-update ADMM check in the report is outside this comparison.

Breast-cancer objective versus iteration



Most of the visible drop happens early.

The curves look flat before the target is met.

Check the gradient sizes in the table.

Raw objective on a linear scale. Looking flat does not mean the stopping check passed.

What the results tell us

What we see	What it means
Same synthetic results	All four pass the gradient check.
Nesterov and FISTA overlap	Their update formulas match here.
Same breast-cancer accuracy	Training can still be unfinished.
ADMM takes longer here	Work inside each update matters.

Choosing a solver for this problem



- **GD:** a simple starting point that works well here.
- **Nesterov:** uses the last movement to look ahead.
- **FISTA:** handles the penalty with a cheap shrink step.
- **ADMM:** separates the problem, but adds inner solves.

Practical conclusion

Compare time at the same training precision. Check the work inside each update.

Limitations and next experiments



- Two small datasets, one split and one timing per method.
- Breast cancer needs more updates to reach the target.
- We have not tested tuned momentum or restarts.

Next experiments

Repeat timings with fixed thread settings. Compare at the same error target, then test other settings and splits.

These runs do not tell us which solver is always fastest.

Conclusion and questions

Main finding

All four solve the synthetic problem, with different amounts of work. On breast cancer, equal test accuracy comes before the training target is met.

LIU Xuyang	26123832r
WANG Zeqing	26029011r
WU Aoqi	26027031r
ZHAO Jixin	26009105r

Thank you. We welcome your questions.

Backup: ADMM residual thresholds

With $\varepsilon = 10^{-6}$ and d parameters:

$$\epsilon_{\text{primal}} = \varepsilon \left(\sqrt{d} + \max(\|w_{k+1}\|_2, \|z_{k+1}\|_2) \right),$$
$$\epsilon_{\text{dual}} = \varepsilon \left(\sqrt{d} + \|\rho u_{k+1}\|_2 \right).$$

The implementation requires all of the following:

$$\|r_{k+1}\|_2 \le \epsilon_{\text{primal}}, \qquad \|s_{k+1}\|_2 \le \epsilon_{\text{dual}}, \qquad \|\nabla F(w_{k+1})\|_2 \le \varepsilon.$$

- The inner Newton solve must also meet its tolerance.
- Metrics evaluate the original $F(w)$ at the returned w.
- ρ is a penalty coefficient, separate from the Newton step size.

Backup: references and code

S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.

A. Beck and M. Teboulle. “A Fast Iterative Shrinkage-Thresholding Algorithm for Linear Inverse Problems.” *SIAM Journal on Imaging Sciences*, 2(1):183–202, 2009.

S. Boyd, N. Parikh, E. Chu, B. Peleato and J. Eckstein. “Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers.” *Foundations and Trends in Machine Learning*, 3(1):1–122, 2011.

Local implementation: `algorithms/`. Shared runner: `experiments/run_compare.py`. Saved numerical evidence: `results/`.